State-Weight Cocycles and Physical Length
Target 57 of the H1 Downstream Programme

Yang–Mills representational dynamics research programme

21 July 2026

Abstract

We prove the scale-dependent state-weight cocycle for changing retained state spaces. Its contraction factors compose exactly and retain the initial root moment. We then distinguish contraction per descriptive RG level from decay per physical distance. A fixed contraction per geometrically growing level is power-law in terminal scale, not a mass gap. The correct bridge is a length-weighted contraction cocycle.

1 Changing-state-space cocycle

Let S_n be finite state spaces, $Q_n : S_n \to S_{n+1}$ positive subkernels, and $V_n : S_n \to [1, \infty)$.

Theorem 1 (Weight cocycle)**.** *If*

$$Q_n V_{n+1} \le q_n V_n, \tag{1}$$

then

$$Q_0 Q_1 \cdots Q_{N-1} V_N \le \left(\prod_{n=0}^{N-1} q_n \right) V_0.$$

For a probability root law ν_0, the total bad-chain mass is at most $\nu_0(V_0) \prod_n q_n$.

Proof. Apply positivity successively to equation (1). Since $1 \le V_N$, the terminal total mass is bounded by the weighted mass. □

The executable six-step star chain has exact mass $2.0603224686684013 \times 10^{-16}$ and upper bound $2.0603224686684008 \times 10^{-16}$, the difference being floating-point roundoff.

2 Physical-length cocycle

Attach a positive physical increment ℓ_n to step n. Define

$$m_N = \frac{-\sum_{n<N} \log q_n}{\sum_{n<N} \ell_n}.$$

Uniform exponential rigidity in physical distance requires

$$\prod_{n<N} q_n \le \exp\left(-m \sum_{n<N} \ell_n \right) \tag{2}$$

with $m > 0$ independent of cutoff, volume, and N.

Proposition 1 (Geometric-level obstruction)**.** *If $q_n = q < 1$ and a terminal scale grows as $R_N = b^N$, then*

$$q^N = R_N^{-\gamma}, \qquad \gamma = -\frac{\log q}{\log b}.$$

Thus a fixed contraction per RG level is naturally power-law in R_N.

For increments $\ell_n = 2^n$ and the H1 value of q, the twenty-level effective mass is only 1.1481773×10^{-4} and tends to zero. In contrast, $q_n = e^{-m\ell_n}$ makes $m_N = m$ exactly; the executable witness gives $m = 0.4$ through total length 1023.

Warning 1. *The cocycle theorem transports representation-dependent rarity. It becomes a mass statement only after the increments are tied to physical separation and after temporal observables are reconstructed densely. Neither identification is supplied by the finite kernel algebra.*

3 Audit

The rectangular cocycle is C2; equation (2) is a C3 conditional bridge. Physical RG kernels, uniform length-scaled contraction, continuum OS construction, and a Hamiltonian gap remain open. Eleven hostile tests cover changing dimensions, root moments, invalid kernels, and both scale laws.